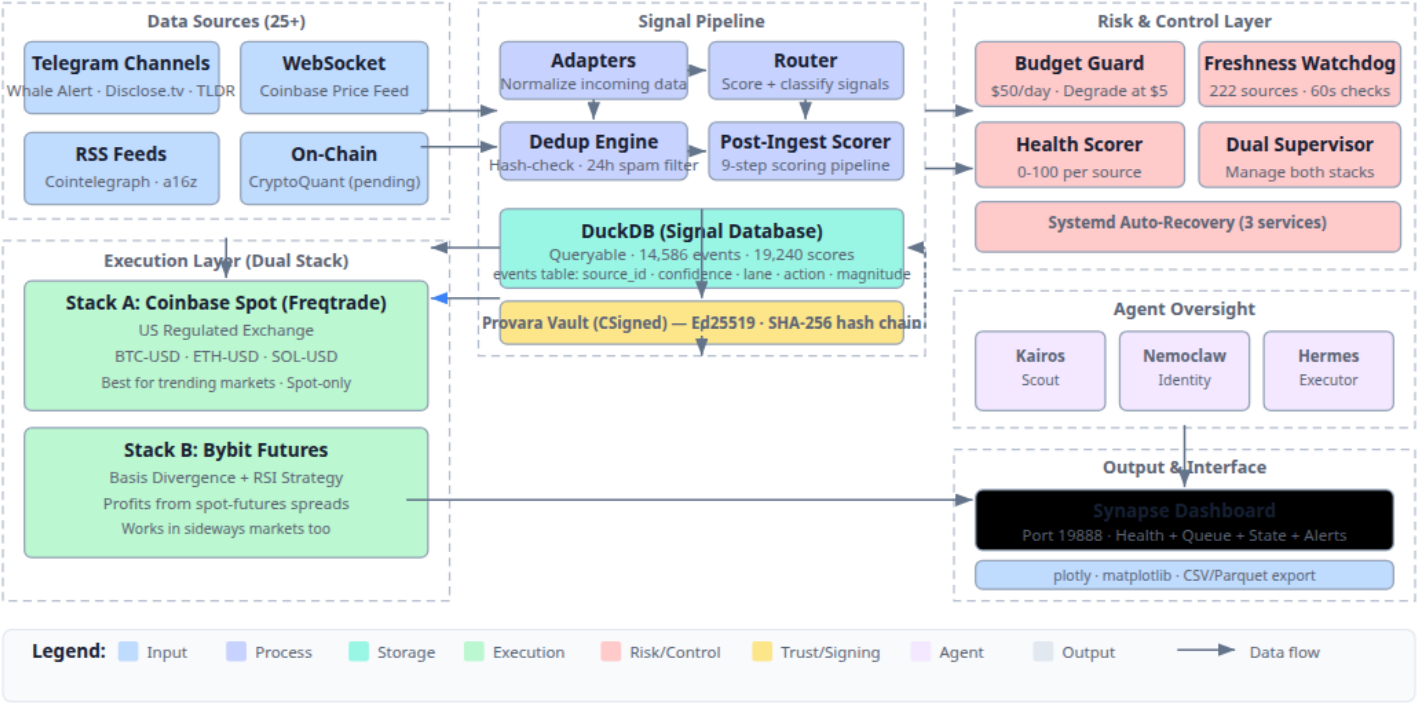


Kestrel Markets

Automated Trading Operating System -- Architecture

Kestrel Markets — Operating System Architecture

Signal Intelligence | Risk-Managed Execution | Cryptographic Audit



How It Works

Step 1: 25+ Data Sources Feed the System

Telegram channels (Whale Alert, Disclose.tv, TLDR), a real-time Coinbase WebSocket, RSS feeds (Cointelegraph, a16z), and on-chain data sources all stream information into the pipeline 24/7. New data arrives every 1-60 seconds.

Step 2: The Pipeline Scores Everything

Adapters normalize incoming data into a standard format. The router assigns a confidence score based on source credibility and event type. The dedup engine removes duplicates (same news from 3 sources = one signal). The post-ingest scorer runs a 9-step evaluation to calculate final signal quality.

Step 3: Signals Go to DuckDB

All scored signals are stored in DuckDB -- a fast embedded SQL database. This holds 14,586+ events, 19,240+ quality scores. The database is queryable and feeds both the execution layer and the dashboard. Every event can also be cryptographically signed into a Provara vault for tamper-evident audit.

Step 4: Dual Execution Stacks Trade

High-confidence signals flow to two independent trading systems: (A) Freqtrade on Coinbase spot exchange -- buys and sells BTC, ETH, SOL on a regulated US market. (B) Bybit futures -- captures profit from price differences between spot and futures markets using basis divergence + RSI strategy. Either stack can trade independently. Both can run simultaneously.

Step 5: Risk Controls Protect Capital

A budget guard caps daily trading at \$50 and enters survival mode at \$5. A freshness watchdog checks all 222 signal sources every 60 seconds. A health scorer rates source quality 0-100. A dual supervisor manages both trading stacks with kill switches. Three systemd services auto-recover from crashes. Every safeguard was built before any trading went live.

Who Runs It

Five specialized AI agents: Kairos scouts for new signals, Nemoclaw manages infrastructure, Hermes executes deployments, Shannon tests for security, and Kestrel coordinates everything. The system runs on a dedicated Ubuntu server. Total daily operating cost: under \$50.

Capital at risk. This document describes an automated trading system under development. Past signal statistics do not guarantee future results. Not financial advice. Hunt Information Systems LLC | June 2026